Ritesh Mamidi

Data & AI Analyst | Applied AI, AI Evaluation, Data & Product Analytics

Austin, TX | riteshmamidi0905@gmail.com

linkedin.com/in/riteshmamidi0905 | github.com/riteshmamidi0905-lab | riteshmamidi0905-lab.github.io

SUMMARY

Data & AI Analyst with 4+ years in data analytics, business analysis and AI-output quality. At Movate, on an Apple Maps client engagement, I validate ML and GenAI outputs, investigate data and model failures, and build QA reporting in Python, SQL and Power BI. In independent projects I build applied AI systems (agent workflows, retrieval, human approval), evaluate them against deterministic baselines, and use the evidence to decide where AI belongs in a business process and where it does not.

SKILLS

Languages & Data: Python, SQL, pandas, scikit-learn, PostgreSQL, Excel

AI & Evaluation: LLM Evaluation, AI Evaluation, Generative AI, AI Agents, Agentic Workflows, RAG, Vector Search, Prompt Engineering, LangGraph, MCP, Benchmark Design, Failure Analysis, Adversarial Testing

Analytics & BI: Power BI, Tableau, ETL, Data Modeling, Data Quality, Data Validation, KPI Design, Forecasting, Statistical Analysis, Hypothesis Testing, A/B Testing, Experimentation, Product Analytics, Funnel & Retention Analysis

Engineering: FastAPI, Pydantic, pgvector, REST APIs, Docker, Git, GitHub Actions (CI), pytest

Business & Product: Business Analysis, Requirements, User Stories, Workflow Analysis, Process Improvement, Stakeholder Communication

PROFESSIONAL EXPERIENCE

Data & AI Analyst | Movate — Apple Maps Client Engagement | Austin, TX Sep 2025 – Present

- Research and validate geospatial data, including points of interest and transit routes, schedules and fares, investigating conflicting and incomplete sources to resolve records accurately.
- Validate ML and GenAI pipeline outputs against policy and quality rubrics; investigate edge cases and systematic failure patterns and document findings for cross-functional partners.
- Build QA/QC checks and operational reporting in Python, SQL and Power BI to support investigation efficiency and reduce operational escalations.
- Apply LLM-assisted analysis and prompt engineering to classify cases and extract operational signals; verify records against APIs.
- Review work in quality-control passes, analyze workflows and support training to improve consistency and process.

Business Analyst | Heart Hospitality LLC | Austin, TX Jan 2024 – Aug 2025

- Automated recurring reporting with Python and SQL ETL pipelines and Power BI/Excel KPI dashboards, reducing manual reporting effort.
- Built forecasting and classification models to support planning.
- Designed relational data models and analyzed operational and customer data in SQL and Python to surface trends that informed business decisions.
- Gathered requirements with stakeholders, documented processes and aligned KPIs and analytics with business goals.

Data Analyst | Liberated Monks Pvt. Ltd. | Hyderabad, India Jan 2022 – Dec 2023

- Built Python and SQL ETL workflows and Tableau dashboards for customer acquisition, engagement and retention analysis.
- Built and evaluated segmentation and propensity (lead-scoring) models for marketing engagement analysis.
- Maintained data validation rules and KPI definitions, translating cross-team requirements into analyses.

SELECTED PROJECTS

Independent, open-source work on synthetic data and tasks; not employer or client work; not deployed.

NASHA AI Lab: Synthetic Hospitality Operating Laboratory

Technologies: Python, FastAPI, PostgreSQL, LangGraph, MCP, RAG, pytest, Docker, GitHub Actions

- Built a simulated hospitality enterprise (synthetic orders, inventory and suppliers; not an operating business) to test where AI belongs in operations: typed tools in permission classes, human approval bound to exact actions, MCP servers behind a trust gateway and a LangGraph agent.
- Compared a deterministic planner with local LLM planners under identical tools and checks: rules passed 259/259 synthetic scenarios vs 5/140 (Qwen2.5-1.5B) and 10/35 (Qwen2.5-7B).

- Tested the tests: 22/22 targeted code faults caught by mutation testing, 11 of 12 injected data-defect types detected, and keyword retrieval (BM25, MRR 0.913) beat neural embeddings (0.672) on a 467-question corpus.
- Built NASHA AI Universe, an interactive headquarters of twelve agent environments that replays hash-verified recordings of governed missions in the browser; reported the failed live model evaluation as failed.

Support Escalation Copilot: Approval-Gated Agentic AI Workflow

Technologies: Python, PostgreSQL, pgvector, RAG, structured LLM outputs

- Built an agentic support workflow for a fictional B2B SaaS team: vector and lexical retrieval with provenance, structured LLM outputs, five typed actions (three human-gated), deterministic policy and PostgreSQL row-level security for tenant isolation.
- Engineered reliability controls: approvals bound to the exact action, idempotent execution with reconciliation of uncertain outcomes, crash recovery and a hash-chained audit log; 489 automated tests and executable tests for 92/92 catalogued attacks.
- Ran a frozen real-model evaluation with a local Qwen3-4B model: 10 of 22 cases reached the expected outcome, failures were at the interface (schema, evidence, parameters), and all four deterministic invariants held in 12/12 injection runs.

LLM & Agent Evaluation Tooling: llmeval and Agent Runtime Benchmark

Technologies: Python, FastAPI, PostgreSQL, Docker

- Built llmeval, an LLM-output evaluation library with deterministic metrics (including faithfulness and hallucination checks), weighted rubrics, hard CI gates and an optional LLM-as-judge backend.
- Benchmarked a local 4B model through a custom agent runtime on a frozen 48-task suite: 43/48 tasks passed and 7/7 runtime controls held in 240/240 runs, yet a permitted read tool leaked a secret into an answer, showing that action safety is not information-flow safety.
- Reported a pre-registered agent-run evaluator (MAREF) as MIXED (33/33 labelled failures detected, 36/154 clean runs flagged) instead of a single favorable score.

EDUCATION

M.S., Business Analytics, St. Francis College, Brooklyn, NY | May 2025 | GPA 3.73

M.Tech (Integrated), Software Engineering, Vellore Institute of Technology (VIT), India | May 2023

CERTIFICATIONS

IBM AI Engineering Professional Certificate, IBM (Coursera) | Jul 2026

Google Advanced Data Analytics Professional Certificate, Google (Coursera) | Apr 2025